Practical No. 12

Web Element Interaction and Synchronization Handling

Aim: Develop a script to automate interaction with textbox, radio button, checkbox, dropdown, alert, and frame elements, using implicit and explicit waits.

Steps

1. Check Java using **java -version** and **javac -version**.
2. Create a Maven project named **SeleniumWebElementPractical** in Eclipse.
3. Add the Selenium dependency to **pom.xml** and update the Maven project.
4. Create **src/test/resources/selenium-demo.html** and add the HTML code below.
5. Create package **com.selenium.practical** and class **WebElementSynchronizationDemo**.
6. Paste the Java code, save, then run as **Java Application**.

Code

pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
         xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
         xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
https://maven.apache.org/xsd/maven-4.0.0.xsd">
    <modelVersion>4.0.0</modelVersion>
    <groupId>com.selenium</groupId>
    <artifactId>SeleniumWebElementPractical</artifactId>
    <version>0.0.1-SNAPSHOT</version>

    <properties>
        <project.build.sourceEncoding>UTF-8</project.build.sourceEncoding>
        <maven.compiler.source>17</maven.compiler.source>
        <maven.compiler.target>17</maven.compiler.target>
    </properties>
    <dependencies>
        <dependency>
            <groupId>org.seleniumhq.selenium</groupId>
            <artifactId>selenium-java</artifactId>
            <version>4.35.0</version >
        </dependency>
    </dependencies>
</project>
```

selenium-demo.html

```
<!DOCTYPE html>
<html>
<head>
    <title>Selenium Web Element Practical</title>
    <style>
        body {
            font-family: Arial;
            margin: 40px;
        }
        .box {
            border: 1px solid #999;
            padding: 20px;
            margin-bottom: 20px;
        }
        iframe {
            width: 500px;
            height: 200px;
        }
    </style>
    <script>
        function showAlert() {
            alert("This is Selenium Alert");
        }
    </script>
</head>
<body>
<h1>Selenium Web Element Practical</h1>
<div class="box">
    <h2>Textbox</h2>
    <label>Name:</label>
    <input type="text" id="name" placeholder="Enter your name">
</div>
```

```
<div class="box">
    <h2>Radio Button</h2>
    <input type="radio" id="male" name="gender" value="Male">
    <label for="male">Male</label>
    <input type="radio" id="female" name="gender" value="Female">
    <label for="female">Female</label>
</div>
<div class="box">
    <h2>Checkbox</h2>
    <input type="checkbox" id="java">
    <label for="java">Java</label>
    <input type="checkbox" id="selenium">
    <label for="selenium">Selenium</label>

</div>
<div class="box">
    <h2>Dropdown</h2>
    <select id="course">
        <option value="">Select Course</option>
        <option value="java">Java</option>
        <option value="python">Python</option>
        <option value="selenium">Selenium</option>
    </select>
</div>
<div class="box">
    <h2>Alert</h2>
    <button id="alertButton" onclick="showAlert()">
        Click for Alert
    </button>
</div>
<div class="box">
    <h2>Frame</h2>
    <iframe id="myFrame"
            srcdoc="
            <html>
            <body>
            <h3 id='frameHeading'>Inside the Frame</h3>
            <input id='frameTextbox' type='text' placeholder='Frame Textbox'>
            </body>
            </html>">
    </iframe>
</div>
</body>
</html>
```

WebElementSynchronizationDemo.java

```
package com.selenium.practical;
import java.io.File;
import java.time.Duration;
import org.openqa.selenium.Alert;
import org.openqa.selenium.By;
import org.openqa.selenium.WebDriver;
import org.openqa.selenium.WebElement;
import org.openqa.selenium.chrome.ChromeDriver;
import org.openqa.selenium.support.ui.ExpectedConditions;
import org.openqa.selenium.support.ui.Select;
import org.openqa.selenium.support.ui.WebDriverWait;
public class WebElementSynchronizationDemo {
    public static void main(String[] args) {
        // Launch Chrome
        WebDriver driver = new ChromeDriver();
        // Maximize browser
        driver.manage().window().maximize();
        // 1. IMPLICIT WAIT
        driver.manage().timeouts()

              .implicitlyWait(Duration.ofSeconds(10));
        // Open local HTML file
        File file = new File(
                "src/test/resources/selenium-demo.html"
        );
        driver.get(file.getAbsolutePath());
        // 2. EXPLICIT WAIT
        WebDriverWait wait =
                new WebDriverWait(driver, Duration.ofSeconds(10));
        // 3. TEXTBOX
        WebElement nameTextbox =
                wait.until(ExpectedConditions
                        .visibilityOfElementLocated(By.id("name")));
        nameTextbox.sendKeys("Rahul");
        System.out.println(
                "Textbox value entered successfully."
        );
        // 4. RADIO BUTTON
        WebElement maleRadio =
                wait.until(ExpectedConditions
                        .elementToBeClickable(By.id("male")));
        if (!maleRadio.isSelected()) {
            maleRadio.click();
        }
        System.out.println(
```

```
                "Radio button selected successfully."
        );
        // 5. CHECKBOX
        WebElement javaCheckbox =
                wait.until(ExpectedConditions
                        .elementToBeClickable(By.id("java")));
        if (!javaCheckbox.isSelected()) {
            javaCheckbox.click();
        }
        System.out.println(
                "Checkbox selected successfully."
        );
        // 6. DROPDOWN
        WebElement courseDropdown =
                wait.until(ExpectedConditions
                        .visibilityOfElementLocated(
                                By.id("course")));
        Select select =
                new Select(courseDropdown);
        select.selectByVisibleText("Selenium");

        System.out.println(
                "Dropdown option selected successfully."
        );
        // 7. ALERT

        WebElement alertButton =
                wait.until(ExpectedConditions
                        .elementToBeClickable(
                                By.id("alertButton")));
        alertButton.click();
        Alert alert =
                wait.until(ExpectedConditions.alertIsPresent());
        System.out.println(
                "Alert message: " + alert.getText()
        );
        alert.accept();
        System.out.println(
                "Alert accepted successfully."
        );
        // 8. FRAME
        WebElement frame =
                wait.until(ExpectedConditions
                        .presenceOfElementLocated(
                                By.id("myFrame")));
        driver.switchTo().frame(frame);
        WebElement frameTextbox =
                wait.until(ExpectedConditions
                        .visibilityOfElementLocated(
                                By.id("frameTextbox")));
        frameTextbox.sendKeys("Text inside frame");
        System.out.println(
                "Textbox inside frame accessed successfully."
        );
        // Switch back to main webpage
        driver.switchTo().defaultContent();
        System.out.println(
                "Returned to main webpage."
        );
        // 9. WAIT AND CLOSE
        try {
            Thread.sleep(2000);
        } catch (InterruptedException e) {
            Thread.currentThread().interrupt();
        }
        driver.quit();
        System.out.println(
                "Browser closed successfully."
        );
    }
}
```